

Developing Strategies to Evaluate Football Plays and to Provide Predictive Schematic Combinations

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Abstract

Football has long been a sport with a high volume of followship and a group that is passionate and loyal. This sport is a key income stream for high school, colleges and on a professional level. This income flow is heavily influenced by winning games. Having a way to identify key variables that affect wins and drives efficiency would valuable assets for a coach when strategizing game plans for a given season. This report takes an analytical approach on using data to expose consistent trends or factors present during a winning period. Various data points were extrapolated from the data, viewed for those most closely correlate to wins. This was successfully done which pointed that this approach is effective and could be applied to different slices of data to possibly predict even more synergies that contribute to wins. This report takes an analytical approach on using data to expose consistent trends or factors present during a winning period. Various data points were extrapolated from the data, viewed for those most closely correlate to wins. This was successfully done which pointed that this approach is effective and could be applied to different slices of data to possibly predict even more synergies that contribute to wins.

1 Introduction

Sports in general are a huge attraction to many. Being a part of sports provides skills like teamwork, discipline, and time management. It also provides or satisfies the urge to compete. Sports provide huge financial gains at all levels – high school, college and professionally. Although many may not admit it, a huge part of this competitive nature ends with a sense of a winner and a loser. In many environments winning is a key driver for many coaches. There's a great deal of skill and thought that goes into crafting, coaching, and executing each play or drive and the most successful teams are those which have a solid playbook and a team of coaches who know when to execute each play. But how does a coach ensure he or she is doing everything within their power to increase their chances of winning? I propose that coaches can break down the plays using analytics. Analytics is a growing area in the information technology sector. Getting data and knowing how to interpret it is key in making decisions as well as being more predictive in decisions that are made.

My aim was to blend the two: take data reported from the college football level, specifically on the offensive side of play and use it to identify trends. Using the data, correlations would be identified as well as any potential combinations of factors that are more inclined to drive future success. By taking these raw data points and deriving critical factors, I was able to find similar patterns across two independent data sets and confirm that the variable was common in both areas and could be stated as a key to higher winning percentages.

2 Related Work

Data Analytics in General

One big area around analyzing data is in the form of automating through machines, specifically computers. This is a much faster approach where literature from (A. Mazighi, L. Ballihi and G. Orhanou 2023; J. Sun 2023) suggests the following things: Security issues are huge concerns faced with data but introducing machine learning or training computers to interpret information allows for the introduction of machine learning algorithms to learn behavior over time and generate results. Essentially, the research discovered that the technology and opportunity existed that “allows the models to make predictions without being instructed on how to do so”. [1] In addition, using big data to generate models in real-time that could ultimately be seen as an intelligent control system. Data analysis is crucial here, but the focus is more on systems having the intelligence to make decisions. [1, 2] This is perhaps an avenue of thought in my study if there are large sets of sports data available. At that point perhaps an algorithm could be used to be more predictive in certain scenarios. This data would have to be collected over time to build a large enough data set to provide a more accurate view but could be beneficial. Considering security and the amount of data available, I would need to consider more manual analytics for the most part to preserve integrity and safeguard details that are more university specific.

Sports Analytics

Continuing further in my research, my next set of reading was to see how analytics was being used in the field of sports. This area has taken hold and is seeing quite a bit of growth in many different areas. Some of these include areas around business operations, health and safety. "The business side of sports includes topics such as ticket pricing, merchandising/brand licensing, venue management, sponsorship return on investment, and TV and radio contracts. For professional sports, it also includes (with inputs from scouts, coaches and owners) financial analytics for player acquisition and trades, and salary cap optimizations." [3] Then there is also a more operational side which is my focus lies. Earlier studies from Cokins, Gary, and Dave Schrader targeted things such as cost-effectiveness, trends on spending and important things to note with health and safety. However, the area that shared some enlightening discovery areas for my search was around the operational side. An interesting aspect of 'space' coverage was investigated. Here “visualizations involve mapping each player's location data (using RFID or GPS or from video annotations) to points on a Cartesian coordinate plane.” [3] Once a sense of location has been obtained, “Derived measures like the distance between players or clusters of players can be calculated. "Cones of coverage" can be computed given windows of time and ball locations, as well as advanced defense statistics never before available.” [3] This is quite intriguing and could be used to evaluate position of players and which plays for example, from the set of football passing plays created the most distance from a defender which then ultimately leads to positive yard gain. [3]

Going further into the latest and greatest breaks in the area, (David Radke and Alexi Orchard. 2023) focused on multiagent systems (MAS) where individuals exhibit teamwork and work within an environment. This study argued that MAS could be used expand development “along two axes: short-term in-game strategy (coaching) and long-term team planning (management).” [4] The focus was placed on the number of players, their positions, how long they were in play, etc. Similar to my thought of evaluating plays, football analytical data would show the players, the yards gained, distance of separation, speed, etc. The MAS system included “tracking sample records features such as timestep, player ID, spatial coordinates, and velocity for each player and the ball or puck, typically amounting to millions of data points in each game.” [4] Another interesting aspect explored in this literature was a way to measure when a player may act on free will and outside of what they have been coached to do. “Learning when this may happen requires a rich understanding of agent and team-level modeling around game theory [30], bounded rationality [44], and group alignment [40] beyond the current state of those fields.” [4] This is another deeper level that may not be explored within my focus but in future more advanced scenarios could very well prove to be a factor that should be looked at more deeply. Each of these are areas that support taking football plays and determining the best and most efficient combinations.

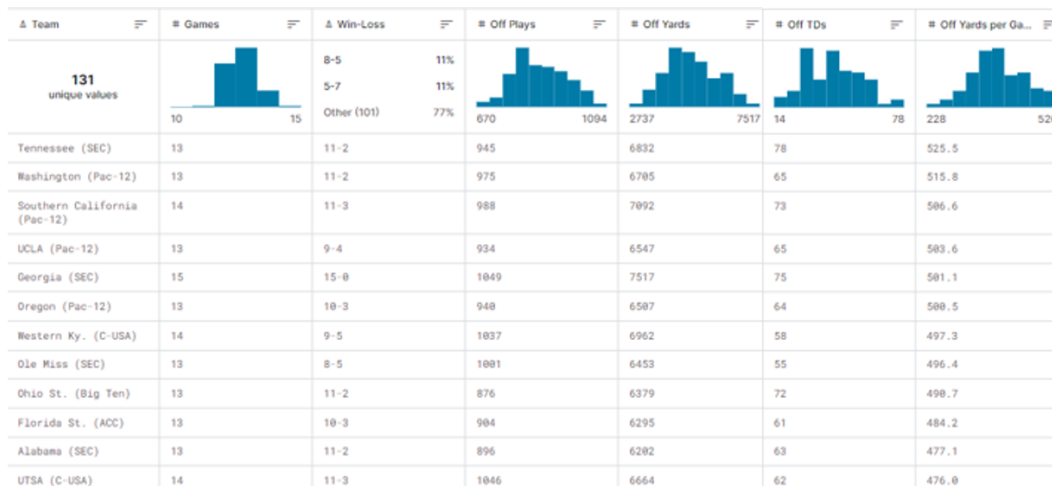
Sports Analytics – Wins and Predictions

My last areas of researching took a more specific course to see what trends and research had been done on taking information and identifying game winners and predictive or futuristic outcomes. Finding useful information in this area could help me to see if it's possible for me to go to do something similar. Two pieces of literature (Bialkowski A., Carr P., Lucey P., Yue Y., Matthews I. 2014; Gifford, Matthew and Bayrak, Tuncay, 2020) focused on wins using things like movement and shots in sports. "“As the game evolved through various rule-changes (i.e., off-side rule) and professionalism (i.e., players could train full-time), so too did formations where coaches/managers set up their team’s formation to best maximize the chances of their team winning while trying to minimize the chances of the opposition”. [5] The first of these had tools to track the ball and player movement using a heat sensor and could somewhat generate some tendencies that were more prevalent at home and how this may differ from strategies away. Factors such as shots taken, possession of the ball, formation, etc. were analyzed. The time possession and formation ideas could possibly be used in my focus when analyzing the data, I have as relates to football formations, schemes, time of possession for each drive and series. The second of these two looked at NFL football and what things can be proven to show a correlation to winning. They were able to take the statistical data points to develop a model for predicting the winner from any match-up. Scatter plots were used to gauge passing and rushing efficiency and logistical binary logic was taken to analyze the likelihood of a field goal. [5, 6] This along with a few other factors were used to help determine who would more than likely end up the winner. This could also be looked at when I seek to pull the data points and see trends forming.

Two other pieces of literature looked more at this area in a slightly different way. The initial one focuses predicting game outcomes (Matt Gifford, Tuncay Bayrak, 2023;) It describes sports analytics as “Sports analytics may be defined as *“the process of using sports-related data to find meaningful patterns and communicate those patterns to help make decisions”*”. [7] From this study, data was gathered to predict patterns and studied to recognize or identify trends. Data collection could be used for predicting wins, player performance and other things such as data management. The next step would be to build a logistical way to structure data and see where patterns form and then construct a confident stand on accurate future outcomes. The second study in this category focuses on the concept of data fusion in sports analytics. (Zerong Jiang and Li Xiao*. 2021) This research and comparative study took data from multiple sources to get a predictive state that when combined would be more solid than any individual or standalone source could be. Therefore, drawing on different aspects of the sports model, more accurate prediction was done. [8] This concept suggests that I could possibly fuse data from a few points, recognize patterns and draw or predict future results in various play patterns.

3 Data

Figure 1: Kaggle data - 20 yr range of FBS stats



Using Kaggle, which is a platform, that serves as a host of data sets that can be used for reporting purposes, data was pulled first against the FBS division. This data spanned across a range of 20 years which would give a strong historical view on which to analyze trends. Data points across aspects such as Offensive plays, Offensive yards, Offensive TDs, PATs, 3rd down conversions, etc. (Figure 1) were collected in various report forms to study. From the many slices of metrics and reporting available, analysis was done to identify any strong correlations.

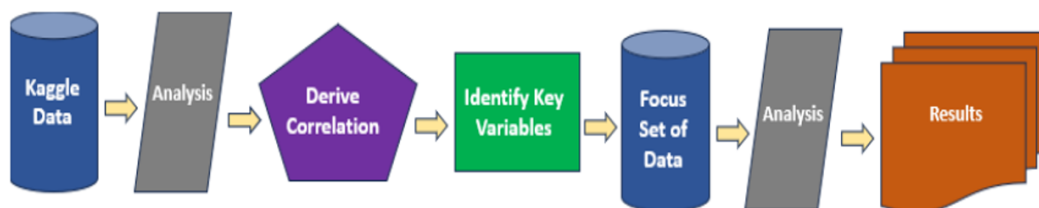
Figure 2: Furman Stats '23-'24 season

	Furman	Mercer	Chattanooga	Western Carolina	YMI	Samford	East Tennessee State	Wofford	Citadel
SCORING									
Total Points Per Game	27.4	24.4	27.5	37.4	15.2	29	19.3	13.8	10
Total Points	356	317	357	411	167	319	212	152	110
Total Touchdowns	43	41	41	53	22	43	27	16	12
1ST DOWNS									
Total 1st downs	251	228	272	288	188	259	154	144	165
Rushing 1st downs	112	103	94	108	81	97	81	65	78
Passing 1st downs	120	100	145	154	88	144	56	60	63
1st downs by penalty	19	25	33	26	19	18	17	19	24
3rd down efficiency	80-198	57-169	63-168	57-128	53-156	68-160	35-140	35-138	52-151
3rd down %	40.4	33.73	37.5	44.53	33.97	42.5	25	25.36	34.44
4th down efficiency	8-19	10-18	4-8	10-22	4-12	11-21	9-15	8-14	13-29
4th down %	42.11	55.56	50	45.46	33.33	52.38	60	57.14	44.83
RECORD vs SOCON	7-1	6-2	6-2	5-3	4-4	4-4	2-6	2-6	0-8

The next steps involved the collection of data from a smaller and more focused target set of data - SoCon football 2023-2024 season. An example of Furman's, one university within the SoCon, dataset is shown in Figure 2. Kaggle did not have reports or data readily available on this division so ESPN's site was used. After collecting the dataset, this information was also sliced into various aspects looking specifically to see if the high correlative factor identified in Step 1 was also having the same impact on this set of data. Lastly some effort was done to see if any additional trends or influencing patterns were present that would help in becoming more predictive in schemes to increase win efficiency.

4 Methods

Figure 3: Pipeline



The method of gathering data began with Kaggle. This site had many graphs, reports, models, etc available for review at the FBS level. This was ideal because the data spanned over 20 years and would show solid trends. This statistical data was downloaded for review. Analysis was performed manually and the strongest correlations were produced. From this key variables with the highest correlations were identified. This would be critical as it would be applied to the next set of studies - focus set being SoCon's 2023-2024 season. Data for this group was analyzed to see if the behaviors held true or if any others were observed. Finally the observations were delivered as result sets.

5 Results

Figure 4: FBS Correlations with Highest Win Percentages

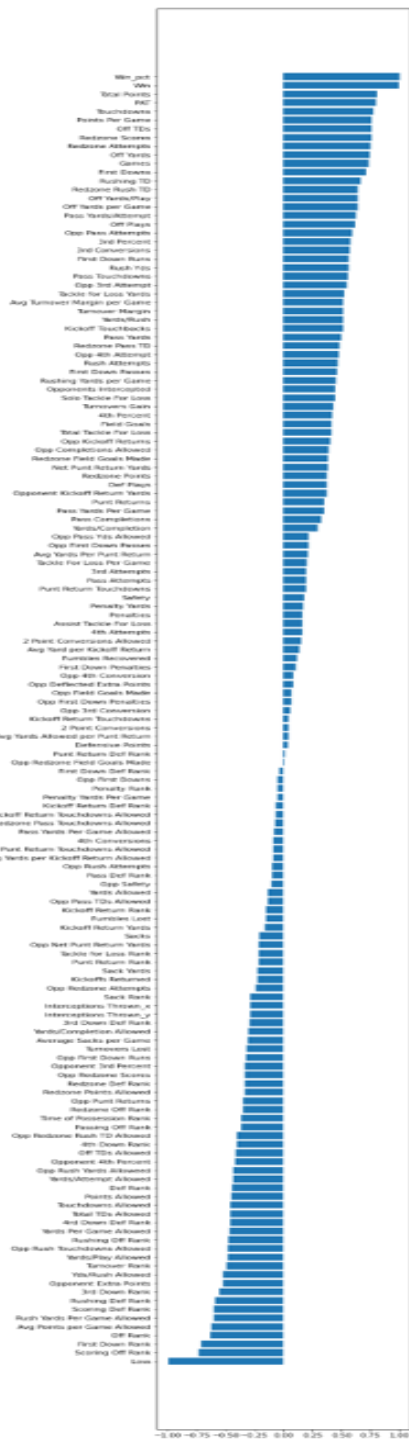
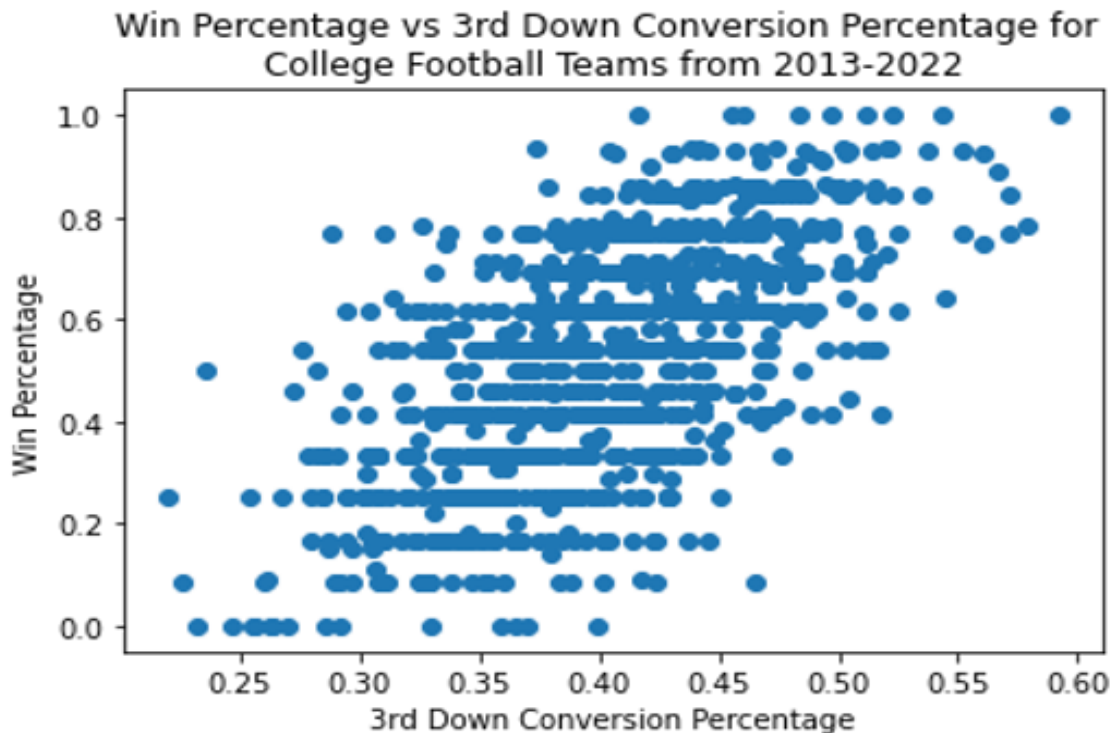


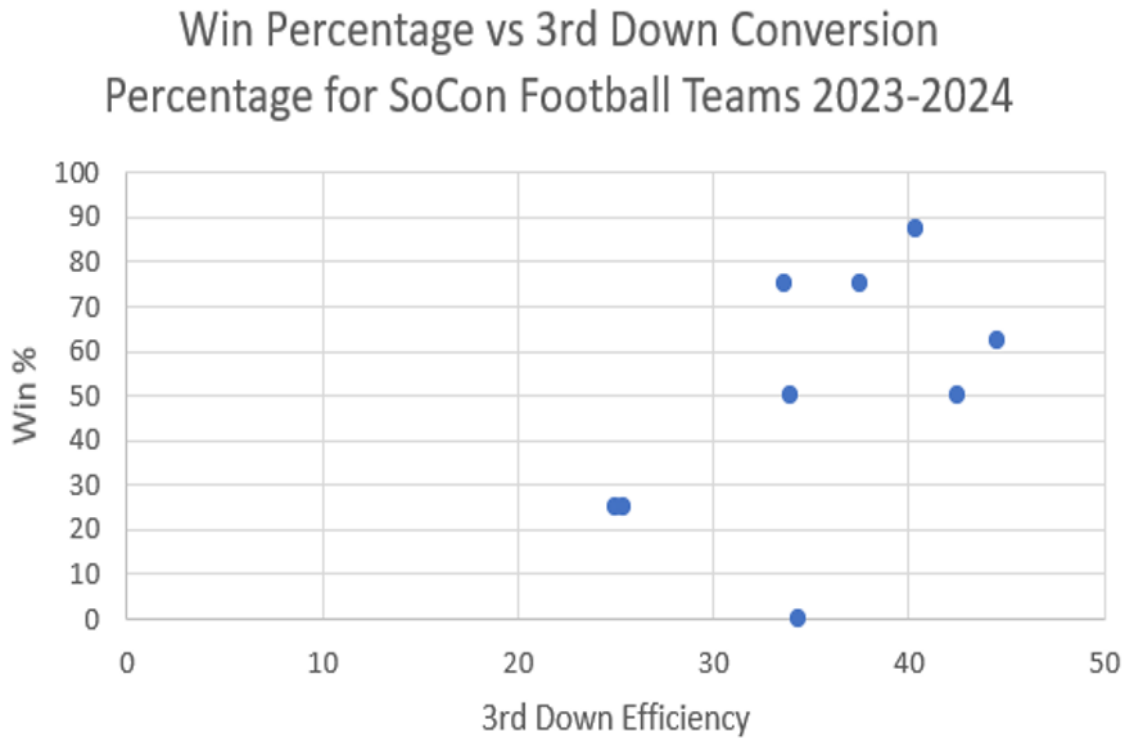
Figure 4 illustrates one of the key sets of facts derived from the Kaggle data set. Considering over 20 years of FBS data had been collected, confidence was high that there was a solid set of teams to inspect and that all of these teams were constrained to use the same basic forms of statistical reporting. After spending quite a bit of time reviewing the information, data for various measures was input in graph form to illustrate their correlation to winning. The correlation results placed a few of the measures very high which indicated they were strong contributors to higher winning efficiencies. For this study, the 3rd down conversion was used for the next level of analysis because of it's placement in the correlation data as well as the fact that this measure is commonly reported in the ESPN stats - which is where the SoCon data would be retrieved.

Figure 5: Win vs. 3rd Down Conversion Percentage for FBS teams



The body of data from Kaggle had 131 unique FBS football teams' data included. There were also approximately 145 different team statistics available. After studying this and identifying those that were high gave more substance to the reliability. Therefore, the fact that consistently higher 3rd down conversion percentages for a given team also trend to have higher overall win percentages, this statistic was used to create a scatter plot to have a supporting visual representation of the trend.

Figure 6: Win vs. 3rd Down Conversion Percentage for SoCon Football Teams 2023-2024



The final result set was generated after applying the identified variable against the target area, which was the SoCon for 2023-2024. Within this set of data there were 9 unique teams and all of which had data reported on ESPN. The statistics from each team was extrapolated and details around the 3rd down percentage efficiency was noted for each. This data, in a similar fashion to the FBS data, was charted in a scatter plot. (Figure 6) Also in similar fashion to the FBS data, the SoCon data showed a high correlation between this variable and higher win percentages.

6 Future Work

In this body of work, I was able to do an analytical dive into areas showing a strong correlation to higher winning percentages. I focused in on one in particular but think that a future study could look at some of the others in the top categories and analyse the data. Also to get a more predictive view on schematic combinations, I feel like having details on specific play calls and outcomes would be beneficial. Lastly, having analytical tools would help to produce the results faster and may be suggestive in variables to consider.

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